

CONTENTS

LIST OF ACRONYMS	i
CHAPTER 1	
INTRODUCTION	10
1.1 INTRODUCTION	
1.2 BACKGROUND OF THE PROJECT	
1.3 PROBLEM STATEMENT	
1.4 OBJECTIVES	
1.5 SCOPE OF THE PROJECT	
CHAPTER 2	
LITERATURE REVIEW	13
2.1 INTRODUCTION	
2.2 EXISTING SYSTEM	
2.3 LIMITATIONS OF EXISTING SYSTEM	
2.4 PROPOSED SYSTEM	

CHAPTER 3 16

METHODOLOGY

3.1 SYSTEM ARCHITECTURE

3.2 COMPONENT DESCRIPTION

3.3 WORKING PRINCIPLE

3.4 FLOWCHART

CHAPTER 4 19

IMPLEMENTATION

4.1 HARDWARE IMPLEMENTATION

4.2 SOFTWARE IMPLEMENTATION

4.3 CIRCUIT DIAGRAM

CHAPTER 5 21

RESULTS AND DISCUSSION

5.1 EXPERIMENTAL SETUP

5.2 RESULTS

5.3 DISCUSSION

CHAPTER 6 23

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

6.2 FUTURE WORK

REFERENCES 26

LIST OF ACRONYMS:

MQ-3 - Alcohol Gas Sensor

GPS - Global Positioning System

GSM - Global System for Mobile Communication

LED - Light Emitting Diode

DC - Direct Current

AC - Alternating Current

IoT - Internet of Things

PCB - Printed Circuit Board

CHAPTER 1 – INTRODUCTION

1.1 INTRODUCTION

Road safety has become a critical global concern due to the increasing number of accidents caused by human negligence, particularly driving under the influence of alcohol. Such accidents not only result in loss of life but also cause severe injuries and economic damage. To address this issue, there is a need for intelligent systems that can actively prevent unsafe driving conditions rather than just responding after an accident occurs.

This project proposes a Smart Safety Helmet system integrated with alcohol detection and engine start prevention mechanisms. The system ensures that a rider is not allowed to start the vehicle if alcohol is detected in their breath. In addition to this preventive feature, the system enhances safety by incorporating accident detection and emergency alert mechanisms using GPS modules. By combining multiple technologies, this project aims to reduce road accidents and improve overall rider safety through smart and automated solutions.

1.2 BACKGROUND OF THE PROJECT

In recent years, the number of road accidents has increased significantly, mainly due to irresponsible driving behaviours such as drunk driving, over-speeding, and lack of safety precautions. Traditional safety measures, such as wearing helmets, provide only passive protection by reducing the severity of injuries during accidents. However, they do not prevent accidents from occurring.

With advancements in embedded systems, sensors, and communication technologies, it is now possible to develop intelligent safety systems that can actively monitor and control driving conditions. The integration of microcontrollers like Esp32, along with sensors and communication

modules, has opened new possibilities in the field of smart transportation and safety.

This project is developed based on these technological advancements, aiming to create a smart helmet that not only protects the rider physically but also prevents unsafe actions such as drunk driving. By combining detection, control, and communication features, the system provides a comprehensive safety solution.

1.3 PROBLEM STATEMENT

Drunk driving is one of the leading causes of road accidents worldwide and poses a serious threat to public safety. Despite strict laws and awareness programs, many individuals continue to drive under the influence of alcohol. Existing safety systems mainly focus on post-accident response, such as sending alerts or providing medical assistance, rather than preventing accidents before they occur.

Moreover, there is a lack of integrated systems that can simultaneously detect alcohol consumption and control vehicle ignition. This gap highlights the need for a reliable and efficient system that can monitor the rider's condition in real time and take immediate action to prevent unsafe driving.

Therefore, the problem addressed in this project is to design and develop a smart system that can detect alcohol levels in the rider's breath and restrict the vehicle ignition accordingly, thereby ensuring enhanced safety and accident prevention.

1.4 OBJECTIVES

The main objectives of this project are as follows:

- To design and develop a smart safety helmet with alcohol detection capability

- To prevent vehicle ignition when alcohol is detected beyond a safe limit
- To implement accident detection using appropriate sensors
- To provide real-time emergency alerts using GPS modules
- To enhance road safety by introducing preventive and automated mechanisms

1.5 SCOPE OF THE PROJECT

The scope of this project is focused on developing a prototype of a smart helmet system that enhances rider safety through detection, prevention, and communication features. The system is primarily designed for two-wheeler riders and demonstrates the concept of ignition control using a relay-based mechanism.

This project serves as a proof-of-concept that can be further extended for real-world vehicle integration. With additional improvements, the system can be implemented in commercial vehicles and integrated with advanced technologies such as IoT, cloud-based monitoring, and mobile applications.

Future enhancements may include features like helmet-wearing detection, real-time data monitoring, and integration with smart transportation systems. Overall, the project has significant potential for practical applications in improving road safety and reducing accident rates.

CHAPTER 2 – LITERATURE REVIEW

2.1 INTRODUCTION

This chapter presents a detailed review of existing technologies and research related to smart helmets, alcohol detection systems, and accident prevention mechanisms. With the rapid advancement of embedded systems and communication technologies, several solutions have been developed to improve road safety. These systems aim to monitor rider behaviour, detect accidents, and provide emergency assistance.

The purpose of this chapter is to analyse such existing solutions, understand their working principles, and identify their strengths and limitations. Based on this analysis, the need for an improved and integrated system is established, which forms the basis for the proposed project.

2.2 EXISTING SYSTEM

Several systems have been developed to enhance road safety, particularly focusing on alcohol detection and accident alert mechanisms. Some vehicles are equipped with alcohol detection devices, such as breath analysers, which prevent the engine from starting if alcohol is detected. Similarly, smart helmets have been designed to detect accidents using vibration or motion sensors and send alerts using GPS and GSM modules.

In addition, certain systems provide features like real-time location tracking, emergency notifications, and helmet usage detection. However, most of these systems operate independently, focusing on either alcohol detection or accident alerting. The lack of integration between these functionalities limits their overall effectiveness in ensuring rider safety.

2.3 LIMITATIONS OF EXISTING SYSTEM

Despite the availability of various safety systems, they suffer from several limitations:

- Lack of integration between alcohol detection and ignition control
- Absence of real-time preventive mechanisms in many systems
- High cost and complexity of implementation
- Limited accessibility and usability for common users
- Focus mainly on detection and alerting rather than prevention
- Dependence on external devices instead of a unified system

These limitations highlight the need for a more efficient and integrated solution that can both detect and prevent unsafe conditions.

2.4 PROPOSED SYSTEM

The proposed system is a smart safety helmet designed to integrate alcohol detection, ignition control, and emergency alert features into a single compact unit. An MQ-3 alcohol sensor is used to detect the presence of alcohol in the rider's breath. The sensor data is processed by an Esp32 microcontroller, which continuously monitors the alcohol level and compares it with a predefined threshold value.

If the detected alcohol level exceeds the threshold, the system activates a relay module to disable the vehicle ignition, thereby preventing the engine from starting. This ensures that the rider cannot operate the vehicle under unsafe conditions, making the system preventive rather than reactive.

In addition to ignition control, the system incorporates GPS modules for real-time location tracking and emergency communication. In case of an accident, the system can automatically send alert messages with location details to predefined contacts. A buzzer is also included to provide immediate audio warnings during critical situations.

This integrated approach enhances overall safety by combining detection, prevention, and emergency response features into a single system. The proposed solution is cost-effective, easy to implement, and suitable for practical applications, making it a reliable system for improving road safety.

CHAPTER 3 – METHODOLOGY

3.1 SYSTEM ARCHITECTURE

The proposed system is designed using a combination of hardware components integrated to perform alcohol detection, ignition control, and emergency alert functions. The main components include an Esp32 microcontroller, MQ-3 alcohol sensor, relay module, GPS module, GSM module, buzzer, and power supply.

The MQ-3 sensor acts as the input device, continuously monitoring the alcohol level in the rider's breath. The sensor output is fed to the Arduino, which serves as the central processing unit of the system. Based on the processed data, the Arduino controls output devices such as the relay module and buzzer.

The relay module is used to simulate and control the vehicle ignition system, while the buzzer provides warning alerts. The GPS module is responsible for obtaining real-time location data, and the Bluetooth is used to transmit alert messages during emergency situations. All components are powered using a regulated power supply, ensuring stable and efficient operation of the system.

3.2 COMPONENT DESCRIPTION

- **Esp32:**
The Esp32 is the main microcontroller used in the system. It processes input data from sensors and controls the output devices based on programmed logic. It acts as the brain of the entire system.
- **MQ-3 Alcohol Sensor:**
The MQ-3 sensor is used to detect the presence and concentration of alcohol in the rider's breath. It provides analog output, which is read by the Arduino for further processing.

- **Relay Module:**
The relay module acts as an electrically controlled switch. It is used to enable or disable the ignition system based on the alcohol detection result, thereby preventing the vehicle from starting when alcohol is detected.
- **GPS Module:**
The GPS module is used to determine the real-time location of the rider. It provides latitude and longitude coordinates, which are useful during emergency situations.
- **Buzzer:**
The buzzer is used to provide audio alerts when alcohol is detected or during abnormal conditions, warning the user instantly.

3.3 WORKING PRINCIPLE

The system operates based on real-time monitoring and decision-making. Initially, the MQ-3 alcohol sensor detects the presence of alcohol from the rider's breath and generates an analogue signal proportional to the alcohol concentration. This signal is continuously read by the Arduino microcontroller.

The Esp32 processes the sensor data and compares it with a predefined threshold value. If the detected alcohol level exceeds the threshold, the system interprets it as an unsafe condition. In such cases, the Arduino keeps the relay module in the OFF state, which breaks the ignition circuit and prevents the vehicle from starting. Simultaneously, the buzzer is activated to provide an immediate warning to the rider.

If the alcohol level is below the threshold, the system allows normal operation by activating the relay, thereby enabling the ignition system. In addition to this, the system is capable of handling emergency situations. In case of an accident, the GPS module retrieves the current location, and the Bluetooth sends an alert message containing this location to predefined

emergency contacts. This ensures timely assistance and improves the chances of rescue.

3.4 FLOWCHART

The working of the system follows a systematic sequence of operations, as represented in the flowchart. Initially, when the system is powered ON, all components including the Esp32, sensors, and communication modules are initialized. The MQ-3 sensor undergoes a brief warm-up period to stabilize its readings and ensure accuracy.

Once initialization is complete, the system continuously monitors the alcohol level using the MQ-3 sensor. The sensor values are read and processed by the Esp32, which compares them with a predefined threshold value. Based on this comparison, a decision is made regarding the safety condition of the rider.

If the detected alcohol level exceeds the threshold, the system identifies it as an unsafe condition. The Esp32 then deactivates the relay module, thereby blocking the ignition system and preventing the vehicle from starting. At the same time, the buzzer is activated to alert the rider.

If the alcohol level is within safe limits, the relay is activated, allowing the ignition system to function normally. Additionally, in case of abnormal conditions such as an accident, the system triggers the alert generation process. The GPS module retrieves the location data, which is then transmitted via the Bluetooth to predefined contacts, ensuring timely emergency response.

CHAPTER 4 – IMPLEMENTATION

4.1 HARDWARE IMPLEMENTATION

The hardware implementation of the proposed system involves the integration of various components such as the Arduino Nano, MQ-3 alcohol sensor, relay module, GPS module, buzzer, and power supply. Each component is connected systematically to ensure proper communication and functionality.

The MQ-3 alcohol sensor is connected to the analogue input pin of the Esp32, allowing it to continuously monitor alcohol levels in the rider's breath. The relay module is connected to a digital output pin of the Esp32 and acts as a switch to control the ignition system. For demonstration purposes, an LED is used to simulate the vehicle ignition, where its ON and OFF states represent engine allowed and engine blocked conditions respectively.

The GPS module is interfaced with the Arduino to obtain real-time location data, while the Bluetooth is used to send alert messages during emergencies. A buzzer is also connected to provide immediate audio warnings when alcohol is detected or when abnormal conditions occur. All components are powered using a regulated power supply to ensure stable and reliable operation of the system.

4.2 SOFTWARE IMPLEMENTATION

The software implementation of the system is carried out using the Arduino Integrated Development Environment (IDE). The program is written in embedded C/C++ and is responsible for controlling the entire system operation.

The code continuously reads the analogue values from the MQ-3 sensor and compares them with a predefined threshold value to determine the presence of alcohol. Based on this comparison, the Arduino controls the relay module

to either allow or block the ignition system. If alcohol is detected above the threshold, the relay is deactivated, preventing the engine from starting, and the buzzer is activated to provide a warning.

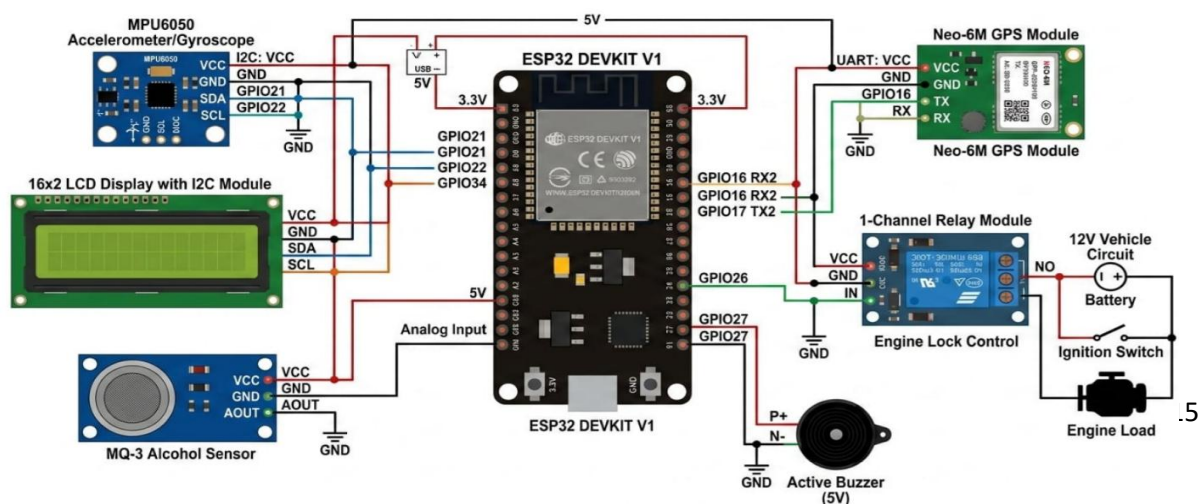
Serial monitoring is used during development and testing to observe sensor readings and calibrate the threshold values for accurate performance. In addition, a mobile application is integrated with the system using Bluetooth connectivity. This application enables communication between the helmet and the user's smartphone, allowing emergency messages to be sent as SMS to predefined contacts along with the exact location obtained from the GPS module.

4.3 CIRCUIT DIAGRAM

The circuit diagram of the system represents the interconnection of all hardware components. The MQ-3 sensor is connected to the analogue input of the Esp32, while the relay module and buzzer are connected to digital output pins. The LED, used to simulate the ignition system, is connected through the relay module to demonstrate switching action.

The GPS modules are interfaced with the Esp32 using serial communication pins. Proper grounding is ensured by connecting all components to a common ground, which is essential for stable operation. Voltage regulation is maintained to supply appropriate voltage levels to each component, preventing damage and ensuring consistent performance. The circuit is designed in a simple and efficient manner, making it easy to implement and troubleshoot. Proper insulation and secure connections are maintained to avoid short circuits and ensure safety during operation.

Smart Vehicle Accident Detection & Alcohol Monitoring System: ESP32 Circuit Diagram



5.1 EXPERIMENTAL SETUP

The developed smart safety helmet system was tested under controlled conditions to evaluate its performance. An LED was used to simulate the vehicle ignition system, where the ON state represents ignition allowed and the OFF state represents ignition blocked. The MQ-3 alcohol sensor was connected to the Arduino to continuously monitor alcohol levels.

To simulate the presence of alcohol, sanitizer vapour was introduced near the sensor, as they contain alcohol content. The relay module was interfaced with the Esp32 to control the LED based on the sensor readings. Additional components such as a buzzer and communication modules (GPS and Bluetooth connectivity) were also integrated to observe alert functionality. The entire setup was powered using a regulated power supply, ensuring stable operation during testing.

5.2 RESULTS

The system was tested under different conditions to verify its functionality:

- **Normal Condition (No Alcohol):**
The MQ-3 sensor detected no significant alcohol presence. The Esp32 activated the relay, allowing current flow, and the LED remained ON, indicating that the engine ignition was enabled.
- **Alcohol Detected Condition:**
When alcohol vapour was introduced near the sensor, the sensor values exceeded the predefined threshold. The Esp32 deactivated the relay, interrupting the circuit, and the LED turned OFF, indicating that the engine ignition was successfully blocked.
- **Alert System:**
When integrated with GPS module and Bluetooth, the system was able to send alert messages containing location details during abnormal conditions such as accidents.

The results confirm that the system responds accurately to varying alcohol levels and performs ignition control effectively.

5.3 DISCUSSION

The experimental results demonstrate that the proposed system successfully detects alcohol and prevents vehicle ignition when unsafe conditions are identified. The relay-based ignition control mechanism proved to be reliable and efficient in simulating real-world operation. The MQ-3 sensor responded effectively to alcohol presence; however, minor calibration was required to set an optimal threshold value for accurate detection.

The system also showed potential for real-time safety applications when integrated with GPS modules for emergency communication. Overall, the system is cost-effective, easy to implement, and suitable for enhancing road safety. With further improvements and real-world integration, the system can be deployed in practical applications to reduce accidents caused by drunk driving.

CHAPTER 6 – CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The project successfully demonstrates the design and implementation of a smart safety helmet capable of preventing vehicle ignition when alcohol is detected. By integrating components such as the MQ-3 alcohol sensor, Esp32 microcontroller, and relay module, the system effectively monitors the rider's condition and ensures that the vehicle cannot be started under unsafe circumstances. This preventive approach helps in reducing the risk of accidents caused by drunk driving.

In addition to ignition control, the system incorporates features such as accident detection and emergency alert mechanisms using GPS modules, which further enhance rider safety. The overall system is cost-effective, easy to implement, and reliable, making it suitable for real-world applications. With further enhancements and integration, this project has the potential to contribute significantly to improving road safety and reducing accident rates.

6.2 FUTURE WORK

Integration with real vehicle ignition system

The system can be directly connected to the actual vehicle ignition circuit, allowing real-time prevention of engine start based on alcohol detection.

IoT-based monitoring

By integrating IoT, the system can send data to cloud platforms for remote monitoring, tracking, and analysis through mobile or web applications.

Advanced sensors for better accuracy

More precise and reliable sensors can be used to improve alcohol detection accuracy and reduce false readings.

Helmet wearing detection system

Additional sensors can be included to ensure that the rider is wearing the helmet before allowing the vehicle to start.

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